

Math 31 Curriculum

Advanced Algebra | Middlesex School Mathematics

COURSE AT A GLANCE

Review Content

REVIEW

Primary resources: [2.2 Use a Problem Solving Strategy](#); [5.2 Properties of Exponents and Scientific Notation](#); [5.1 Add and Subtract Polynomials](#); [5.3 Multiply Polynomials](#)

UNIT LEARNING OBJECTIVES

I can calculate a percentage, percent increase, or percent decrease.

I can simplify an expression using exponent properties.

I can use polynomial vocabulary to distinguish terms, factors, coefficients, degree, and polynomial structure.

I can expand a product of two binomials.

Linear Functions and Function Notation

REQUIRED

Primary resources: [College Algebra 2e 3.1 Functions and Function Notation](#); [College Algebra 2e 3.3 Rates of Change and Behavior of Graphs](#); [College Algebra 2e 3.2 Domain and Range](#); [College Algebra 2e 4.1 Linear Functions](#); [Intermediate Algebra 2e 4.1 Solve Systems of Linear Equations with Two Variables](#)

UNIT LEARNING OBJECTIVES

I can interpret function notation as a statement about input and output.

I can determine whether a graph or table represents a function.

I can evaluate a function or solve for an input using a graph, table, or formula.

I can identify independent and dependent variables in a relationship.

I can calculate average rate of change from two points.

I can interpret interval notation and identify increasing or decreasing intervals.

I can determine total change from an average rate of change over an interval.

I can determine and interpret the units of an average rate of change.

I can identify a linear function from a table, graph, or formula.

I can interpret the slope and intercept of a linear model.

I can determine the intersection of two linear relationships graphically or algebraically.

I can write equations of horizontal, vertical, parallel, and perpendicular lines through a specified point.

Function Behavior and Structure

REQUIRED

Primary resources: [3.1 Functions and Function Notation](#); [3.2 Domain and Range](#); [3.5 Transformation of Functions](#); [3.4 Composition of Functions](#); [3.7 Inverse Functions](#); [3.3 Rates of Change and Behavior of Graphs](#)

UNIT LEARNING OBJECTIVES

I can determine whether a relation is a function.

I can evaluate a function from a formula, graph, or table.

I can determine domain and range from a formula, graph, or table.

I can determine algebraic domain restrictions caused by denominators and even roots.

I can graph a piecewise function from its formula.

I can write a piecewise formula that represents a graph.

I can graph a transformed function using shifts, stretches, compressions, and reflections.

I can describe how a transformation affects domain, range, or asymptotes.

I can evaluate a composition from formulas, graphs, or tables.

I can write a formula for a composition of two functions.

I can find an inverse function algebraically.

I can evaluate and interpret an inverse from a graph, formula, or table.

I can determine and interpret the units of an inverse function.

I can explain how the domain and range of a function relate to those of its inverse.

I can determine the range of a function by analyzing the domain of its inverse.

I can verify algebraically that two functions are inverses.

I can interpret the meaning and units of a composition.

I can classify a graph or table as concave up or concave down using changes in average rate of change.

Function Behavior and Structure (Continued)

REQUIRED

UNIT LEARNING OBJECTIVES - CONTINUED

I can determine whether a function is invertible using the horizontal line test.

I can graph an inverse function as a reflection across the line $y = x$.

Quadratic Functions and Equations

REQUIRED

Primary resources: [5.1 Quadratic Functions](#); [2.5 Quadratic Equations](#)
Supplemental resources: [5.3 Graphs of Polynomial Functions](#); [5.5 Zeros of Polynomial Functions](#)

UNIT LEARNING OBJECTIVES

I can identify the vertex, intercepts, axis of symmetry, and concavity of a quadratic function.

I can connect standard, factored, and vertex forms to features of a quadratic graph.

I can rewrite a quadratic function in a form that reveals a requested feature.

I can solve a quadratic equation by factoring or using the quadratic formula.

I can graph a quadratic function using its algebraic structure and key features.

I can construct a quadratic function from specified zeros, a vertex, or graphical information.

I can create and interpret a quadratic trajectory model.

Exponential Functions and Models

REQUIRED

Primary resources: [6.1 Exponential Functions](#); [6.2 Graphs of Exponential Functions](#)

UNIT LEARNING OBJECTIVES

I can distinguish linear change from exponential change using differences, ratios, or percent change.

I can identify initial value and growth or decay factor in an exponential model.

I can write an exponential function from a table, graph, or context.

I can evaluate and interpret an exponential model.

I can calculate growth under periodic compounding.

I can calculate growth under continuous compounding.

I can determine an exponential formula from two points or other sufficient data.

I can determine domain, range, intercepts, and asymptotes of an exponential function.

I can compare the growth rates of two exponential functions.

I can determine the percent growth or decay represented by a continuous exponential model.

Logarithmic Functions and Models

REQUIRED

Primary resources: [6.3 Logarithmic Functions](#); [6.5 Logarithmic Properties](#); [6.6 Exponential and Logarithmic Equations](#); [6.7 Exponential and Logarithmic Models](#); [6.4 Graphs of Logarithmic Functions](#)
Supplemental resources: [6.2 Graphs of Exponential Functions](#)

UNIT LEARNING OBJECTIVES

I can convert between logarithmic and exponential forms.

I can evaluate a logarithm using its exponential meaning.

I can expand or condense an expression using product, quotient, and power properties.

I can solve an exponential equation using logarithms.

I can solve a logarithmic equation using exponential form and check domain restrictions.

I can calculate and interpret doubling time or half-life.

I can determine domain, range, intercepts, and asymptotes of a logarithmic function.

I can explain how exponential and logarithmic functions undo one another.

I can solve a multi-step exponential or logarithmic equation.

I can determine an intersection of exponential functions graphically or algebraically.

I can rewrite a discrete exponential model as an equivalent continuous exponential model.

I can solve and interpret an exponential or logarithmic modeling problem.

Extension Content

EXTENSION

Primary resources: [3.6 Absolute Value Functions](#); [3.5 Transformation of Functions](#); [6.7 Exponential and Logarithmic Models](#)

UNIT LEARNING OBJECTIVES

I can analyze an absolute-value function using transformations.

I can determine even or odd symmetry from a graph or formula.

I can interpret a contextual logarithmic scale.